

BCSE302P – DATABASE SYSTEMS LAB
SOCIETAL DIGITAL INNOVATION PROJECT

Theme:

1. **Identify a Real Societal Problem**
2. **Invent a Digital Solution**
3. **Build It Using Database Technology**

IMPORTANT ANNOUNCEMENT

This project forms part of the 30-mark Database Systems Laboratory component. Students must complete the project development before the final Project Expo/Challenge. The contest/expo is planned for October 2026. The exact date and venue will be announced later. An internal project review carrying 10 marks will be conducted before the final event.

1. Purpose of the Project

Students are expected to identify a genuine societal problem, study existing practices and digital solutions, identify a gap, propose an innovative digital solution, design and implement a database-centric system, develop a functional prototype, validate it using realistic scenarios, and demonstrate potential societal impact.

2. Core Project Philosophy

Start with: “What real problem exists, why is the current solution insufficient, and what new digital solution can I create?”

3. Open Innovation – No Fixed Project Titles

Students are NOT restricted to predetermined application titles. The following domains are only challenge areas to stimulate problem discovery. Teams may identify their own significant societal problem within these areas or propose another relevant problem, subject to faculty approval.

Track	Challenge Domain	Example Areas
T1	Safety & Security	Personal safety, child safety, elderly safety, public safety
T2	Emergency & Disaster Resilience	Floods, accidents, cyclones, evacuation, rescue
T3	Water & Natural Resources	Water scarcity, groundwater, lakes, irrigation, water quality
T4	Agriculture & Rural Development	Farmers, equipment, crops, markets, storage, rural services
T5	Waste & Circular Economy	Waste, recycling, reuse, repair, surplus resources
T6	Healthcare & Well-being	Healthcare access, preventive support, elderly care, medical resources
T7	Inclusion & Accessibility	Persons with disabilities, elderly and vulnerable communities
T8	Environment & Climate	Coastal issues, pollution, biodiversity, climate resilience

T9	Education & Human Development	Skills, learning resources, employability, peer learning
T10	Smart Community & Civic Life	Local services, public infrastructure, community resources
T11	Mobility & Urban Living	Transport, parking, pedestrian safety, last-mile connectivity
T12	Livelihood & Social Entrepreneurship	Local skills, artisans, micro-businesses, resource sharing

4. Mandatory Problem-Discovery Requirements

- Identify a real and clearly defined societal problem.
- Identify who is affected and where the problem occurs.
- Collect supporting evidence through observation, interviews, surveys, literature, reports or credible data where feasible.
- Study existing non-digital and digital solutions.
- Identify limitations/gaps in existing solutions.
- Define the proposed innovation and explain why it is different.
- Identify intended users/stakeholders and potential beneficiaries.
- Explain how the proposed system can create measurable societal benefit.

5. Innovation / Novelty Requirement

The project should not simply reproduce an already available application. Teams must demonstrate a meaningful gap and a clearly defined innovation.

- Existing solution(s): What already exists?
- Limitation: What important need remains unmet?
- Proposed innovation: What exactly is new?
- Expected improvement: Why could the proposed solution be more useful?

6. Database Systems – Mandatory Component

The database must be the central technical component. The project will be evaluated as a Database Systems project, not merely as a mobile/web application.

- Requirements and data specification
- ER/EER diagram with meaningful entities, attributes and relationships
- Relational schema and constraints
- Normalization with justification
- Database implementation and realistic data
- DDL, DML, complex SQL, joins, nested/aggregate queries and views where appropriate
- Indexes and query-performance considerations
- Transactions and concurrency considerations where applicable
- Triggers, procedures/functions or database programming where justified
- Role-based access/security and data integrity where applicable
- Application–database integration
- Database-driven reports, dashboards or analytics

7. Technology Freedom

Teams may use AI/ML, IoT, GIS/spatial data, computer vision, NLP, APIs, cloud services, recommendation systems, optimization or NoSQL where these technologies genuinely solve the identified problem. Technology must not be added merely to make the project appear advanced.

8. Expected Development Outcome

- A functional [web/mobile/desktop application, platform, dashboard or integrated digital prototype](#).
- A working database with realistic or carefully designed test data.
- An end-to-end workflow from user/stakeholder input to database processing and useful output.
- Testing using realistic scenarios.
- A clear explanation of potential societal impact.
- A roadmap for improvement, scalability and deployment.

9. Recommended Innovation Maturity Levels

Level	Description	Expected Evidence
1	Database Application	Database + functional application
2	Intelligent Database Application	Database + analytics/decision support
3	Smart Digital Solution	Database + justified AI/ML/IoT/GIS/etc.
4	Societal Innovation Prototype	Working prototype + realistic validation + measurable impact
5	Startup/TRL Candidate	Validated prototype + scalability + stakeholder/user evidence + deployment/business/social model

10. TRL Expectation

The recommended target is approximately **TRL 4–5**, depending on the project. TRL indicates technology/prototype maturity; it is not a measure of societal importance.

- TRL 3 – Proof of concept demonstrated.
- TRL 4 – Working prototype validated in a controlled/laboratory environment.
- TRL 5 – Prototype validated using realistic data, users, locations or scenarios.
- Higher TRLs may be pursued where feasible, but external deployment is not mandatory.

11. Project Team

- Maximum team/batch size: 3 students.
- Every member must have a clearly identifiable contribution.
- All members should participate in problem analysis, database design, development, testing and presentation.

12. Mandatory Deliverables

No.	Deliverable	Expected Content
1	Problem Discovery Report	Problem, beneficiaries, evidence, existing solutions and gap
2	Innovation Proposal	Proposed solution, novelty and expected societal impact
3	Software Requirements	Functional/non-functional and stakeholder requirements

4	Database Design	ER/EER, relational schema, normalization and constraints
5	Application Prototype	Working digital solution integrated with database
6	Testing & Validation	Test cases, realistic scenarios and results
7	Impact & TRL Report	Societal impact, scalability and TRL justification
8	Expo Presentation	Demo, poster/pitch and project explanation

13. Internal Review – 10 Marks

An internal project review carrying 10 marks will be conducted before the final contest/expo. Teams must show sufficient progress toward a complete, database-centric and demonstrable prototype.

Review Criterion	Marks
Problem understanding & evidence	2
Innovation/novelty and existing-solution gap	2
Database design and implementation progress	2
Application/prototype progress	2
Testing, societal impact and TRL readiness	2

14. Final Project Expo / Challenge – October 2026

The Project Expo/Challenge is planned for October 2026. The exact date, venue, schedule and contest procedure will be announced separately. Students must complete development and validation before the event.

- Teams will demonstrate the working prototype.
- Judges will evaluate technical/database implementation, innovation, societal relevance, prototype maturity and presentation.
- Strong projects may be encouraged for further development towards higher TRL, startup/innovation opportunities, external competitions or other suitable avenues, subject to eligibility and applicable rules.

15. Suggested 30-Mark Structure

Component	Marks	Focus
Internal Review	10	Problem, innovation, database and prototype progress
Project Expo – Technical & Database Evaluation	10	Database implementation, application integration, testing and demonstration
Project Expo – Innovation, Societal Impact & TRL	10	Novelty, impact, scalability, prototype maturity and presentation

16. Evaluation Expectations

- A strong user interface alone will not receive high marks.
- Copied applications without meaningful innovation will not be considered strong.
- The database must be demonstrably central to the solution.
- Students must be able to explain schema, queries, constraints, transactions and design decisions.
- Novelty claims must be supported by an existing-solution study.
- Societal impact should be explained using measurable or defensible outcomes.
- Projects should be designed with scalability and real-world deployment in mind.

Project Journey

DISCOVER - RESEARCH - IDENTIFY GAP - INNOVATE - MODEL DATABASE - DEVELOP - TEST - VALIDATE - MEASURE IMPACT - REVIEW(10) - EXPO/CHALLENGE (OCT-2026)

Student Preparation Checklist

- Team of maximum 3 students formed.
- Societal problem identified and approved.
- Stakeholders/beneficiaries identified.
- Existing solutions studied.
- Innovation/gap clearly documented.
- Requirements specification prepared.
- ER/EER model completed.
- Relational schema and normalization completed.
- Database implemented and populated.
- SQL and relevant database features implemented.
- Application connected to database.
- Testing and validation completed.
- Societal impact and TRL evidence prepared.
- Prototype ready for internal review and final Expo/Challenge.

Message to Students

Do not think of this as just another laboratory application. Think of it as an opportunity to identify a problem that affects real people, challenge the way it is currently solved, and build something genuinely useful. You are free to explore your own idea, domain and technology. The strongest projects will combine a meaningful societal problem, genuine innovation, strong Database Systems implementation, a functional prototype and evidence of potential real-world impact.